
Memorization on Fine-Tuned LLMs

Parjanya Prashant
(PID: A15778224)
pprashant@ucsd.edu

Aldan Creo
(PID: A69034625)
acreo@ucsd.edu

Jonah Yi
(PID: A17040180)
jwy010@ucsd.edu

Sarthak Kala
(PID: A69034307)
skala@ucsd.edu

Abstract

Fine-tuning large language models using Low-Rank Adaptation (LoRA) has become widespread due to its parameter efficiency, yet its memorization characteristics remain poorly understood. We conduct a systematic study of memorization in LoRA fine-tuning, examining how two key hyperparameters (rank and training epochs) affect both syntactic memorization and privacy vulnerabilities. Through comprehensive experiments, we measure verbatim sequence memorization and quantify privacy risks using membership inference attacks across varying LoRA configurations. Our analysis reveals that memorization increases significantly with the number of epochs but is not greatly affected by the rank. We evaluate existing mitigation strategies, focusing on the Goldfish loss approach for reducing memorization. Finally, we propose enhanced mitigation techniques that effectively reduce both syntactic memorization and membership inference vulnerability while maintaining task performance.

1 Introduction

Large language models serve as foundation models with broad general capabilities [Team et al., 2023, OpenAI, 2025, Dubey et al., 2024, Team et al., 2025] and are often fine-tuned for specific downstream tasks [Howard and Ruder, 2018]. Full fine-tuning is computationally expensive and memory intensive, leading to widespread adoption of parameter-efficient fine-tuning methods, most notably Low-Rank Adaptation (LoRA) [Hu et al., 2022]. LoRA decomposes weight updates into two low-rank matrices that are trained while keeping the base model frozen, drastically reducing memory requirements and training costs. These LoRA adapters are frequently released by companies and organizations as standalone modules for public use [Mangrulkar et al., 2022], enabling practitioners to adapt foundation models without access to significant computational resources.

However, fine-tuning induces memorization and privacy risks, particularly when models are trained on private or proprietary data. We focus on two primary attack vectors. First, extraction attacks enable adversaries to recover training data verbatim or near-verbatim through carefully crafted prompts, raising concerns about copyright infringement and plagiarism [Carlini et al., 2021]. Second, membership inference attacks (MIAs) allow attackers to determine whether specific data points were part of the training set, violating privacy guarantees [Shokri et al., 2017, Mattern et al., 2023]. While memorization has been extensively studied in the pretraining setting, it remains underexplored for fine-tuning. Existing fine-tuning work [Mireshghallah et al., 2022b] considers only membership inference attacks without examining extraction vulnerabilities.

Recently, there has been significant interest in studying properties of fine-tuning (both supervised fine-tuning and reinforcement learning) with respect to LoRA rank [Schulman and Lab, 2025]. We extend this line of work by investigating how memorization varies with LoRA rank during supervised fine-tuning. In this paper, we study how both extraction and privacy risks change with LoRA rank across different training epochs. Our experiments show that memorization generally increases with training duration, but LoRA rank does not significantly affect memorization behavior. Additionally, numerous mitigation methods have been proposed [Hans et al., 2024, Prashant et al.], most targeting syntactic

memorization (extraction risk) rather than privacy. We evaluate the Goldfish loss function and quantify its mitigation effectiveness. Finally, we propose two improvements to Goldfish, collectively termed Surgeonfish, that substantially enhance memorization mitigation performance.

2 Literature Review

Memorization: LLMs have been shown to verbatim or near-verbatim reproduce parts of their training data [Carlini et al., 2021, 2022, Karamolegkou et al., 2023], often through prefix attacks where specific prompts elicit memorized sequences. Memorization can also be formalized through the lens of adversarial compression [Schwarzschild et al., 2024]. Empirically, memorization scales with model size [Biderman et al., 2023, Carlini et al., 2022, Zhou et al., 2023], as larger models possess greater capacity to store training examples. Recent work has established that models exhibit a maximum memorization capacity that scales linearly with parameter count [Morris et al., 2025]. MIAs have been used as a proxy for memorization [Miresghallah et al., 2022a] and have been studied with respect to finetuning too [Miresghallah et al., 2022b].

Mitigating memorization: Mitigation strategies can be categorized into three approaches: pre-training, unlearning, and inference-time methods. Pre-training methods include deduplication [Kandpal et al., 2022] and Goldfish [Hans et al., 2024], though these are computationally expensive and can degrade performance. Unlearning methods [Maini et al., 2023, Sakarvadia et al., 2024] similarly suffer from performance drops. Inference-time approaches include MemFree [Ippolito et al., 2022], which blocks exact matches, CPFuse [Abad et al., 2024], which fuses models trained on disjoint data, ParaPo [Chen et al., 2025], which uses DPO to prefer paraphrased outputs, and TokenSwap [Prashant et al.], which combines logits with an auxiliary model to disrupt memorization. However, inference-time methods remain vulnerable to determined attackers who can still extract training data through adversarial techniques.

3 Method

3.1 Finetuning

We finetune models using Low-Rank Adaptation (LoRA), which updates pretrained weights W_0 via low-rank decomposition. The adapted weight becomes

$$W = W_0 + BA$$

where $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ with rank $r \ll \min(d, k)$. Only the low-rank matrices A and B are trained while W_0 remains frozen, reducing trainable parameters significantly. We train each configuration for 10 epochs to study how memorization evolves over training duration.

3.2 Mitigation

Goldfish Goldfish loss [Hans et al., 2024] mitigates syntactic memorization by excluding tokens from loss computation. For a mask $G \in \{0, 1\}^L$ and sequence of length L , the loss is

$$\mathcal{L}_{\text{goldfish}} = -\frac{1}{|G|} \sum_{i=1}^L G_i \log P(x_i | x_{<i})$$

where $G_i = 0$ drops token i . This prevents learning complete verbatim sequences while maintaining task performance.

Surgeonfish We propose Surgeonfish, which trains on the most likely predicted token that is not the ground truth. For each position i , let $\hat{x}_i = \arg \max_{x \neq x_i} P(x | x_{<i})$ be the top non-ground-truth prediction. The loss becomes

$$\mathcal{L}_{\text{surgeon}} = -\frac{1}{L} \sum_{i=1}^L \log P(\hat{x}_i | x_{<i})$$

This leverages pretrained capabilities to learn patterns without memorizing exact sequences.

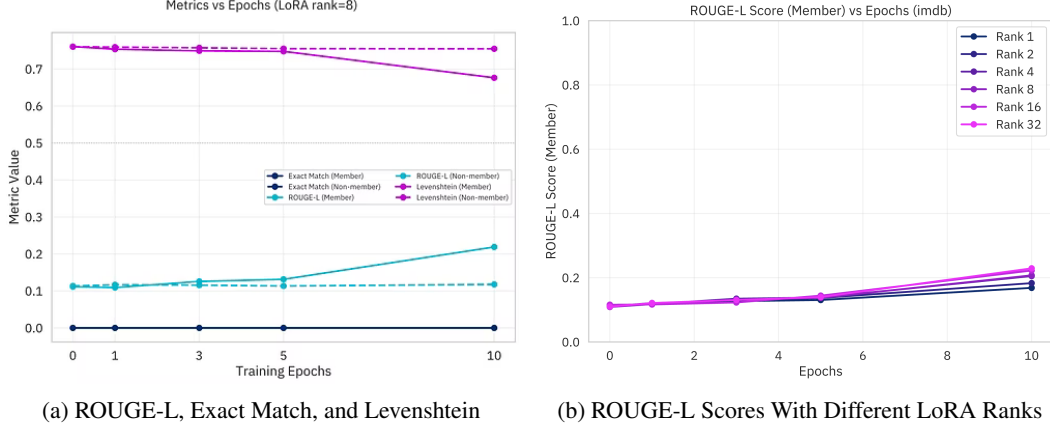


Figure 1: Memorization Metrics Across Different Training Epochs

Surgeonfish+ Surgeonfish+ applies the technique selectively to low-entropy tokens. We define entropy $H_i = -\sum_x P(x|x_{<i}) \log P(x|x_{<i})$ and threshold τ . The loss is

$$\mathcal{L}_{\text{surgeon+}} = -\frac{1}{L} \sum_{i=1}^L \log P(y_i|x_{<i}), \quad y_i = \begin{cases} \hat{x}_i & \text{if } H_i < \tau \\ x_i & \text{otherwise} \end{cases}$$

This focuses mitigation on memorization-prone tokens while preserving learning signal elsewhere.

4 Experiments

We empirically evaluate how LoRA fine-tuning influences memorization under varying training durations, LoRA ranks, and mitigation strategies. Our study examines two distinct dimensions of memorization: (1) *syntactic memorization*, referring to near-verbatim reproduction of training sequences, and (2) *privacy leakage*, captured via membership inference attacks (MIAs). Experiments are conducted on three text generation datasets—WritingPrompts, IMDB, and Wikitext—which differ in size, style, and repetitiveness.

4.1 Experimental Setup

We fine-tune a pretrained decoder-only LLM using LoRA ranks $r \in \{1, 2, 4, 8, 16, 32\}$, training each model configuration for 1–10 epochs to observe both early-stage and overfitting behavior. For evaluation, we measure Exact Match, ROUGE-L, and normalized Levenshtein distance, which quantify syntactic reproduction, as well as naïve and shadow-model MIAs, which assess privacy leakage. We additionally report the perplexity gap between member and non-member sequences, a known proxy for memorization and overfitting.

4.2 Syntactic Memorization

Across all datasets and LoRA ranks, syntactic memorization remains low. As seen in Figure 1, ROUGE-L, Exact Match, and Levenshtein similarity show minimal increase for the first several epochs and only modest growth by epoch 10. The absence of substantial near-verbatim reproduction suggests that LoRA updates provide limited additional capacity for explicit memorization.

Furthermore, LoRA rank does not meaningfully influence memorization. Metrics for ranks 1 through 32 are nearly indistinguishable, indicating that increasing the dimensionality of the low-rank update does not amplify the model’s ability to store training sequences. This stands in contrast to classical parameter-count scaling laws observed in full-model pretraining.

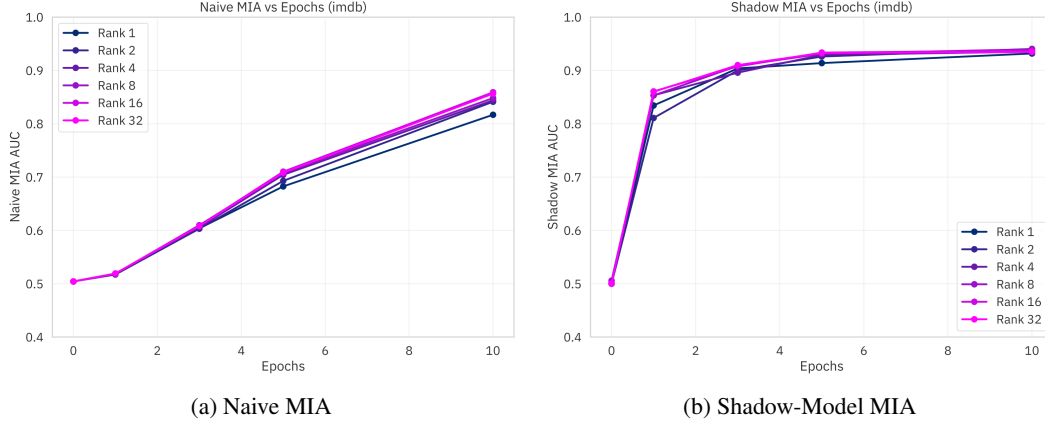


Figure 2: Naive and Shadow-Model MIA

4.3 Membership Inference Attacks

While syntactic memorization is weak, MIAs reveal significant privacy leakage, as seen in Figure 2. The naïve MIA exhibits a steadily increasing AUC, approaching 0.80 by epoch 10. Shadow-model MIAs—known to better approximate adversarial inference—achieve AUC values above 0.90 by epoch 3 and saturate near 0.95, demonstrating that membership leakage arises long before any surface-level memorization is detectable.

Strikingly, MIA performance remains invariant across LoRA ranks, reinforcing that memorization risk is driven predominantly by training duration and overfitting rather than LoRA capacity.

4.4 Perplexity Gap Analysis

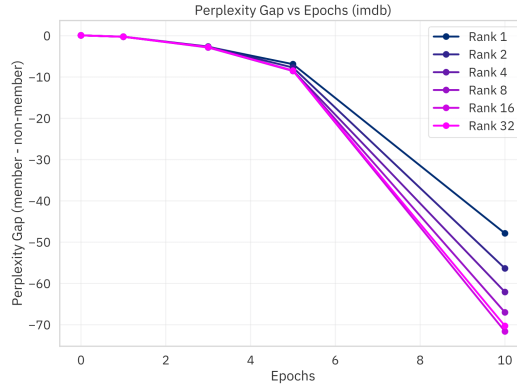


Figure 3: Perplexity Gap

We observe an increasingly negative perplexity gap (member perplexity $<$ non-member perplexity) as training progresses. In figure 3 we can see that by epoch 10, member perplexity can be over 60 points lower than non-member perplexity. This widening gap closely mirrors MIA performance and highlights that confidence overfitting, not verbatim copying, drives privacy vulnerabilities during LoRA fine-tuning.

4.5 Mitigation Evaluation

We evaluate Goldfish, Surgeonfish, and Surgeonfish+ as syntactic memorization mitigation methods in Figure 4. All approaches reduce ROUGE-L relative to baseline training, with Surgeonfish+ yielding the largest reduction, particularly for longer training schedules. These results support the hypothesis

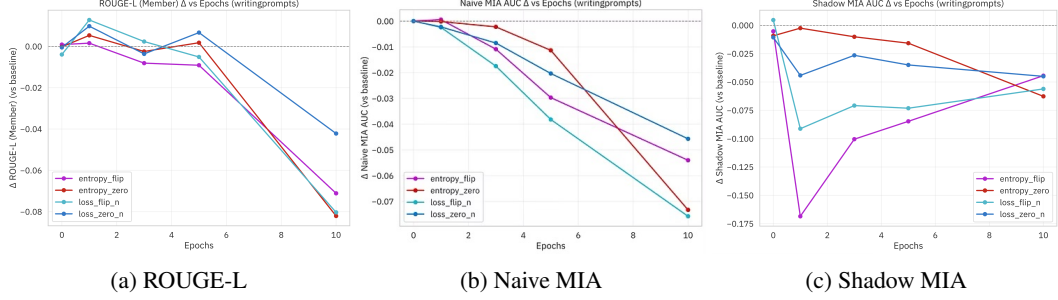


Figure 4: Syntactic Memorization Mitigation Methods' Affect on Memorization

that low-entropy tokens disproportionately contribute to memorization and benefit from selective perturbation.

For naïve MIAs, mitigations reduce AUC slightly (0.01–0.07), but shadow MIAs remain largely unaffected. This indicates that syntactic-focused mitigation is insufficient for countering distributional privacy leakage, which originates from confidence calibration rather than exact sequence reproduction.

Overall, our results show that:

- LoRA fine-tuning produces very limited syntactic memorization.
- Strong MIAs succeed even in the absence of verbatim reproduction.
- Training duration—not LoRA rank—is the primary factor influencing memorization.
- Syntactic mitigation reduces near-verbatim copying but fails to meaningfully reduce shadow-model MIAs.

Limitations and Future Work

While our study provides a detailed characterization of memorization during LoRA fine-tuning, several limitations warrant further investigation. First, our experiments focus on a single model architecture and scale. Memorization behavior may differ substantially for larger models or encoder–decoder architectures.

Second, the mitigation strategies evaluated—Goldfish, Surgeonfish, and Surgeonfish+ primarily target syntactic memorization and do not address privacy (they are not robust against MIAs). Their limited impact on shadow-model MIAs demonstrates the need for mitigation methods explicitly designed for privacy preservation, such as differentially private fine-tuning.

Finally, Surgeonfish+ requires a manually selected entropy threshold, which may not generalize across datasets or models. Future work should explore adaptive thresholding.

AI Usage Statement

We used AI tools to help with developing our code and editing the paper. However, all research ideas, experimental design, implementation decisions, initial drafts and core contributions are our own work.

Contributions Statement

Item	Aldan Creo	Jonah Yi	Parjanya Prashant	Sarthak Kala
Ideation	✓	✓	✓	✓
Literature Review			✓	
Experiments	✓			
Midterm Report Writing		✓		✓
Final Report Writing		✓	✓	✓
Presentation	✓	✓	✓	✓

References

- Javier Abad, Konstantin Donhauser, Francesco Pinto, and Fanny Yang. Copyright-protected language generation via adaptive model fusion. *arXiv preprint arXiv:2412.06619*, 2024.
- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling, 2023. URL <https://arxiv.org/abs/2304.01373>.
- Nicholas Carlini, Florian Tramer, Eric Wallace, Matthew Jagielski, Ariel Herbert-Voss, Katherine Lee, Adam Roberts, Tom Brown, Dawn Song, Ulfar Erlingsson, et al. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pages 2633–2650, 2021.
- Nicholas Carlini, Daphne Ippolito, Matthew Jagielski, Katherine Lee, Florian Tramer, and Chiyuan Zhang. Quantifying memorization across neural language models. *arXiv preprint arXiv:2202.07646*, 2022.
- Tong Chen, Faeze Brahman, Jiacheng Liu, Niloofar Mireshghallah, Weijia Shi, Pang Wei Koh, Luke Zettlemoyer, and Hannaneh Hajishirzi. Parapo: Aligning language models to reduce verbatim reproduction of pre-training data. *arXiv preprint arXiv:2504.14452*, 2025.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Abhimanyu Hans, Yuxin Wen, Neel Jain, John Kirchenbauer, Hamid Kazemi, Prajwal Singhania, Siddharth Singh, Gowthami Somepalli, Jonas Geiping, Abhinav Bhatele, et al. Be like a goldfish, don’t memorize! mitigating memorization in generative llms. *arXiv preprint arXiv:2406.10209*, 2024.
- Jeremy Howard and Sebastian Ruder. Universal language model fine-tuning for text classification. *arXiv preprint arXiv:1801.06146*, 2018.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022.
- Daphne Ippolito, Florian Tramèr, Milad Nasr, Chiyuan Zhang, Matthew Jagielski, Katherine Lee, Christopher A Choquette-Choo, and Nicholas Carlini. Preventing verbatim memorization in language models gives a false sense of privacy. *arXiv preprint arXiv:2210.17546*, 2022.
- Nikhil Kandpal, Eric Wallace, and Colin Raffel. Deduplicating training data mitigates privacy risks in language models. In *International Conference on Machine Learning*, pages 10697–10711. PMLR, 2022.
- Antonia Karamolegkou, Jiaang Li, Li Zhou, and Anders Søgaard. Copyright violations and large language models. *arXiv preprint arXiv:2310.13771*, 2023.
- Pratyush Maini, Michael C Mozer, Hanie Sedghi, Zachary C Lipton, J Zico Kolter, and Chiyuan Zhang. Can neural network memorization be localized? *arXiv preprint arXiv:2307.09542*, 2023.
- Sourab Mangrulkar, Sylvain Gugger, Lysandre Debut, Younes Belkada, Sayak Paul, Benjamin Bossan, and Marian Tietz. PEFT: State-of-the-art parameter-efficient fine-tuning methods. <https://github.com/huggingface/peft>, 2022.
- Justus Mattern, Fatemehsadat Mireshghallah, Zhijing Jin, Bernhard Schölkopf, Mrinmaya Sachan, and Taylor Berg-Kirkpatrick. Membership inference attacks against language models via neighbourhood comparison. *arXiv preprint arXiv:2305.18462*, 2023.
- Fatemehsadat Mireshghallah, Kartik Goyal, Archit Uniyal, Taylor Berg-Kirkpatrick, and Reza Shokri. Quantifying privacy risks of masked language models using membership inference attacks. *arXiv preprint arXiv:2203.03929*, 2022a.

- Fatemehsadat Mireshghallah, Mohammadali Naseri, Nils Holzenberger, Pratyush Mani, Harsha Nori Ramaswamy, Mohammad Khani, Daniel Tran, and Florian Tramer. Memorization in nlp fine-tuning methods. *arXiv preprint arXiv:2205.12506*, 2022b.
- John X Morris, Chawin Sitawarin, Chuan Guo, Narine Kokhlikyan, G Edward Suh, Alexander M Rush, Kamalika Chaudhuri, and Saeed Mahloujifar. How much do language models memorize? *arXiv preprint arXiv:2505.24832*, 2025.
- OpenAI. Gpt-5 system card. Technical report, OpenAI, August 13 2025. URL <https://cdn.openai.com/gpt-5-system-card.pdf>. PDF.
- Parjanya Prajakta Prashant, Kaustubh Ponkshe, and Babak Salimi. Tokenswap: A lightweight method to disrupt memorized sequences in llms. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*.
- Mansi Sakarvadia, Aswathy Ajith, Arham Khan, Nathaniel Hudson, Caleb Geniesse, Kyle Chard, Yaoqing Yang, Ian Foster, and Michael W Mahoney. Mitigating memorization in language models. *arXiv preprint arXiv:2410.02159*, 2024.
- John Schulman and Thinking Machines Lab. Lora without regret. *Thinking Machines Lab: Connectionism*, 2025. doi: 10.64434/tml.20250929. <https://thinkingmachines.ai/blog/lora/>.
- Avi Schwarzschild, Zhili Feng, Pratyush Maini, Zachary C Lipton, and J Zico Kolter. Rethinking llm memorization through the lens of adversarial compression. *arXiv preprint arXiv:2404.15146*, 2024.
- Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE symposium on security and privacy (SP)*, pages 3–18. IEEE, 2017.
- Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.
- Gemma Team, Aishwarya Kamath, Johan Ferret, Shreya Pathak, Nino Vieillard, Ramona Merhej, Sarah Perrin, Tatiana Matejovicova, Alexandre Ramé, Morgane Rivière, et al. Gemma 3 technical report. *arXiv preprint arXiv:2503.19786*, 2025.
- Zhenhong Zhou, Jiuyang Xiang, Chaomeng Chen, and Sen Su. Quantifying and analyzing entity-level memorization in large language models, 2023. URL <https://arxiv.org/abs/2308.15727>.